Matching Compressor

User Guide

1. Introduction

Matching Compressor is a multi-knee compressor/expander (up to three knees) that allows you to match the dynamic characteristics of one audio signal to another.

If you have a reference audio signal with your desired dynamic profile and another signal you want to shape similarly, the plugin can analyze both and determine compressor/expander parameters that best match the reference.

Key features:

- Standard set of parameters: Gain, Threshold, Ratio, Knee Width, Attack, Release, along with Envelope Type and Stereo Processing mode
- Up to three compressor/expander knees
- Automatic matching of dynamic parameters
- Support for both audio files and input buses (main and sidechain) as sources for reference and destination audio
- Graphical representation of the processing curve

Matching Compressor is distributed as a VST plugin for Windows and Linux. A MacOS version (VST and AU) is also available, but it is not signed or notarized; use it at your own risk.

The source code of the plugin is available at:
https://github.com/justonem0reuser/MatchingCompressor.

2. Working with Matching Compressor

2.1. Main Window

2.1.1. Elements Description

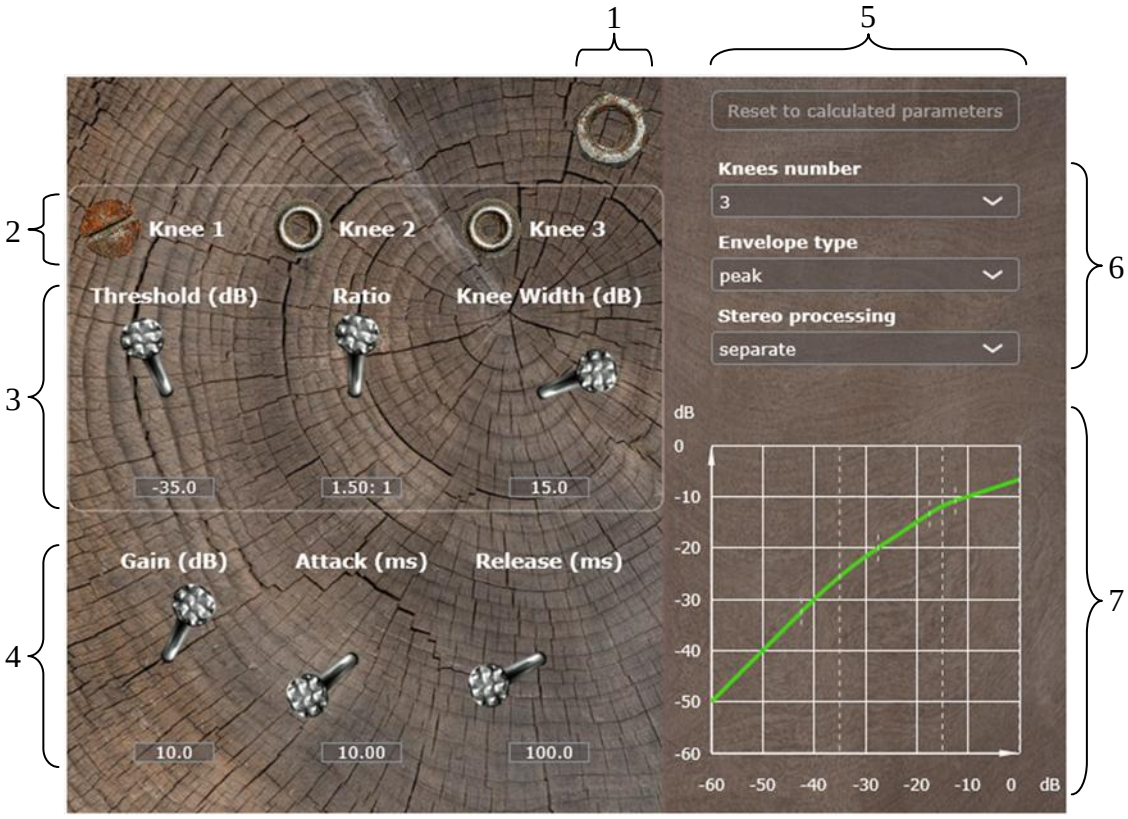

Fig.1. Main window

The main window contains the following elements:

1. Matching button - Opens the matching window.

2. Knee selection buttons – Two or three radio buttons for selecting the active knee (hidden if only one knee is used).

3. Knee-related sliders:

- **Threshold** - Sets the signal level at which compression or expansion begins.
- **Ratio** - Determines how strongly the signal is reduced or increased once it crosses the threshold.
- **Knee Width** - Controls how gradually processing is applied around the threshold.

4. Chain-related sliders:

- **Gain** - Adjusts the overall output level after processing.
- **Attack** - Sets how quickly the processor responds after the signal exceeds the threshold.

- **Release** - Controls how quickly the processor returns to normal after the signal falls below the threshold.

5. **Reset to calculated parameters** button - Resets all sliders and combo boxes to their default or calculated values (disabled until matching is performed).

6. Plugin-related combo boxes:

- **Knees number** - Sets the number of knees in the processing chain (1, 2 or 3).
- **Envelope Type** - Selects whether detection is based on **peak** levels (fast, precise) or **RMS** levels (smoother, average-based).
- **Stereo Processing** - Defines how stereo channels are processed (no effect on mono audio): independently (**separate**), linked by the louder channel (**max**), or linked by the average level (**mean**).

7. Plot - Displays the compressor curve.

2.1.2. Working With Sliders And Knees

Doubl-clicking a slider resets it to its default state.

Before the first matching operation, the default values are:

- **Threshold**: 0 dB
- **Knee Width**: 0 dB
- **Ratio**: 1:1
- **Gain**: 0 dB

These correspond to no processing with a hard knee. Default **Attack** and **Release** values are 10 ms and 100 ms, respectively.

After matching, default values correspond to the calculated results (for **Threshold**, **Knee Width**, **Ratio**, **Gain**) or fixed values (for **Attack** and **Release**).

If the **Ratio** is less than 1:1, expansion is applied instead of compression.

Selecting a knee using the radio buttons links the knee-related sliders to that knee.

If multiple knees are used then knees cannot overlap and are kept in ascending order. The maximum threshold is 0 dB. Therefore, if the upper boundary of a knee (threshold + half of its width) reaches 0 dB, adding another knee becomes impossible (the corresponding option is disabled). Moreover, available ranges of the **Threshold** and **Knee Width** sliders are limited by these constraints.

2.1.3. Plot

The plot uses a standard representation for dynamic processors: the X-axis shows the input level in dB, and the Y-axis shows the output level in dB.

The thick green curve shows the current settings. The thin white curve shows the parameters obtained during matching (if available).

Vertical dashed lines indicate each knee's threshold, and shorter dashed lines indicate knee boundaries.

2.2. Matching Window

To open the matching window, click the Matching button.

2.2.1. Elements Description

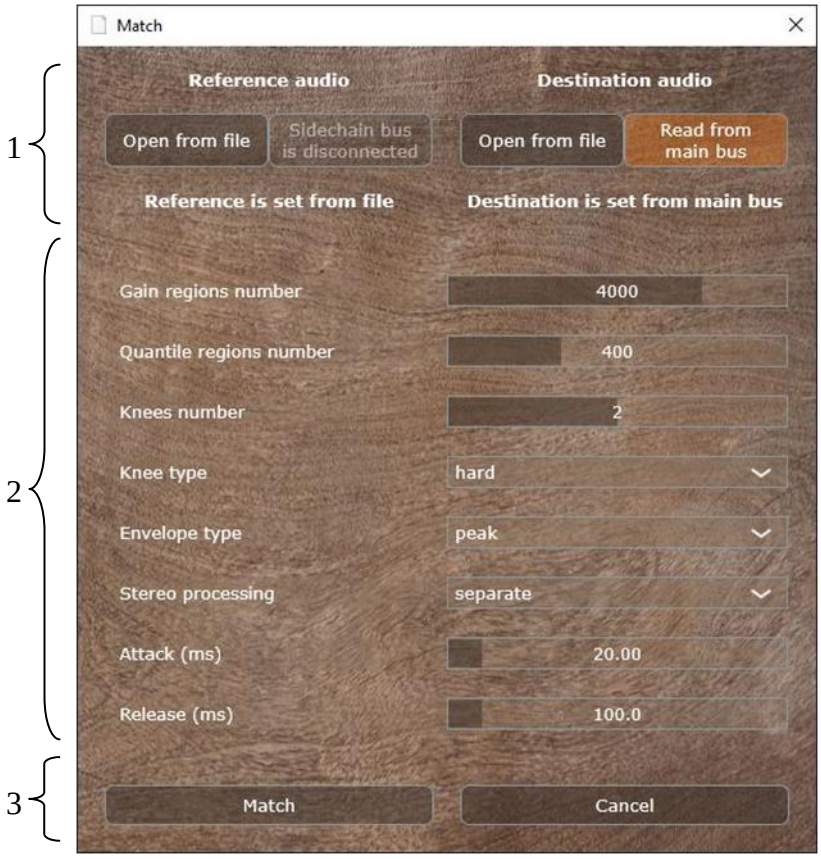

Fig.2. Settings window

This window contains:

1. Reference and destination audio panel.

2. Matching settings.

3. **Match** and **Cancel** buttons.

2.2.2. Setting Reference And Destination Audio

Reference audio is the signal with desired dynamics. **Destination audio** is the signal to be processed. Both can be loaded from files or recorded from buses.

From files:

- Click **Open from file** button.
- Select a file from a dialog.
- A confirmation message will be displayed.

From buses:

- Ensure that the required bus is connected via your DAW. The destination audio is read from the main bus that is always connected with the processed track. The reference audio is read from the sidechain bus that is usually disconnected, so you should connect it with the right track if needed; otherwise the related button is disabled, so you will not be able to set the reference audio from the bus.
- Stop playback.
- Choose which of the audios (reference, destination or both) should be set from buses by clicking appropriate buttons: **Read from sidechain bus** or/and **Read from main bus**. The clicked buttons change their color.
- Start playback from the desired point. The two buttons become disabled until stop.
- Stop playback after the desired fragment.
- A confirmation message will be displayed.

The **Match** button becomes available only after both signals are set.

2.2.3. Matching settings and process

There are two types of settings: algorithm parameters and fixed parameters.

The **Gain regions number** and **Quantile regions number** are algorithm parameters (see **Maching Compressor Methods Description** document for further information). Higher values increase accuracy but require more processing time. It is recommended to set the **Gain regions number** several times higher than the **Quantile regions number**.

Knee type is the third algorithm parameter. The corresponding combo box has two options. If the **Hard Knee** option is selected, knee widths are not determined and are set to 0 dB instead. The **Soft Knee** option means that knee widths are determined.

Knees number, **Envelope Type**, **Stereo Processing**, **Attack**, and **Release** are parameters that are fixed during the estimation process.

After clicking the **Match** button, the estimation process starts. Once it is finished, the matching window closes and the processing parameters are set:

- **Threshold**, **Ratio**, **Knee Width** for each knee, as well as the overall **Gain**, are set based on the calculation results.
- **Knees number**, **Envelope Type**, **Stereo Processing**, **Attack**, **Release** are applied from the matching window.

3. Tips

- The processed audio will sound more similar to the reference if both have similar frequency content. Consider using an EQ before this plugin (if it aligns with your processing goals). Plugins that allow adjusting the amplitude-frequency response of the processed audio to match the reference can be helpful in this regard. Choosing a fragment in the same musical style as the reference can lead to a better result.
- Setting attack and release times to zero is not recommended for any compressor. The plugin does not attempt to estimate optimal attack and release values, they must be specified manually.
- Since matching involves optimization of a non-linear function, convergence to a global minimum is not guaranteed. Increasing the number of knees (especially when using **Soft Knee** option) does not always yield more accurate results.
- Accurate knee width estimation requires higher region counts. Using **Hard Knee** allows you to use lower region counts and, therefore, reduce computation time with minimal loss of accuracy.